

August 21, 2026

1 Sharp second-order causal accounting

The first-order causal theorem determines the asymptotic rate

$$C_\infty^{\text{causal}}(P, Q) = \max\{H(P), H(Q)\}$$

for two finite actions. The remaining finite-horizon question is whether the sublinear residual contains a canonical second-order term.

This section gives an exact answer in the equal-entropy case.

Throughout, logarithms are base 2. Zero-mass symbols are deleted from the alphabets. For a finite probability law P , write

$$\iota_P(x) := -\log_2 P(x), \quad h_P := H(P),$$

and

$$V_P := \text{Var}_{X \sim P}(\iota_P(X)), \quad \sigma_P := \sqrt{V_P}.$$

We assume

$$H(P) = H(Q) = h.$$

Define the causal residual

$$E_n := C_n^{\text{causal}}(P, Q) - nh.$$

The main result is the following.

Theorem 1.1 (Sharp causal second-order law). *Let P, Q be nondegenerate finite probability distributions satisfying*

$$H(P) = H(Q) = h.$$

Then

$$C_n^{\text{causal}}(P, Q) = nh + \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| \sqrt{n} + o(\sqrt{n}).$$

Equivalently,

$$\lim_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} = \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q|.$$

The proof has two independent parts.

The lower bound uses the fact that a single exact causal seed must support *every* adaptive policy simultaneously.

The matching upper bound constructs one exact one-way causal coupling whose information-spectrum fluctuations are asymptotically coupled monotonically and whose continuous interval state is then exactized by a prefix-free fair-bit transcript at only logarithmic boundary cost.

1.1 The adaptive seed converse

Let S denote any discrete seed defining an exact horizon- n causal response tree.

For a deterministic adaptive policy π , let

$$Y^\pi(S) = (Y_1^\pi(S), \dots, Y_n^\pi(S))$$

be the realized output trajectory.

If

$$a_t^\pi(y^{t-1})$$

is the action selected by the policy after history y^{t-1} , define the trajectory probability

$$p_\pi(y^n) := \prod_{t=1}^n P_{a_t^\pi(y^{t-1})}(y_t)$$

and its self-information

$$I_\pi(y^n) := -\log_2 p_\pi(y^n).$$

Because all actions have entropy h , write

$$M_{\pi,n} := I_\pi(Y^\pi) - nh.$$

Lemma 1.2 (Finite-policy seed-profile converse). *For every finite family Π of deterministic adaptive policies,*

$$\boxed{H(S) \geq nh + \mathbb{E} \max_{\pi \in \Pi} M_{\pi,n}.}$$

Proof. Let s be an atom of the seed law, with

$$\Pr(S = s) = \gamma_s.$$

For each deterministic policy π , the atom s produces one definite trajectory

$$y^\pi(s).$$

Exactness implies that the total seed mass mapped to this trajectory is exactly

$$p_\pi(y^\pi(s)).$$

Since the particular atom s is one term in that mass,

$$\gamma_s \leq p_\pi(y^\pi(s))$$

for every π .

Therefore

$$-\log_2 \gamma_s \geq I_\pi(y^\pi(s))$$

simultaneously for every policy, and hence

$$-\log_2 \gamma_s \geq \max_{\pi \in \Pi} I_\pi(y^\pi(s)).$$

Average over S :

$$H(S) = \mathbb{E}[-\log_2 \gamma_S] \geq \mathbb{E} \max_{\pi \in \Pi} I_\pi(Y^\pi).$$

Since all policies have the same mean trajectory entropy nh ,

$$I_\pi(Y^\pi) = nh + M_{\pi,n},$$

which proves the claim. □

1.2 Threshold policies and the diffusion limit

Assume first that

$$\sigma_P \geq \sigma_Q.$$

Set

$$a := \sigma_P, \quad b := \sigma_Q.$$

If $a = b$, the lower bound in Theorem 1.1 is zero and there is nothing to prove in this subsection. We therefore suppose

$$a > b.$$

For each $z \in \mathbb{R}$ and horizon n , define the threshold policy $\pi_{z,n}$ recursively as follows. Let

$$M_0 = 0.$$

At time t , select the action with standard deviation a if

$$M_{t-1} < z\sqrt{n},$$

and select the action with standard deviation b if

$$M_{t-1} \geq z\sqrt{n}.$$

After observing the next output, update

$$M_t = M_{t-1} + \imath_{A_t}(Y_t) - h.$$

Thus (M_t) is a martingale with bounded increments and

$$\mathbb{E}[(M_t - M_{t-1})^2 \mid \mathcal{F}_{t-1}] = \begin{cases} a^2, & M_{t-1} < z\sqrt{n}, \\ b^2, & M_{t-1} \geq z\sqrt{n}. \end{cases}$$

The following lemma supplies the diffusion limit required below. It is included explicitly because a simple oscillating-random-walk invariance theorem does not by itself cover arbitrary finite surprisal increments.

Lemma 1.3 (Threshold martingale diffusion approximation). *For fixed $z \in \mathbb{R}$, define*

$$X_n(t) := \frac{M_{\lfloor nt \rfloor}}{\sqrt{n}}, \quad 0 \leq t \leq 1.$$

Then

$$X_n \Longrightarrow X^{(z)}$$

in $D([0, 1])$, where $X^{(z)}$ is the unique weak solution of

$$dX_t = \sigma_z(X_t) dB_t, \quad X_0 = 0,$$

with

$$\sigma_z(x) = \begin{cases} a, & x < z, \\ b, & x \geq z. \end{cases}$$

In particular,

$$\frac{M_n}{\sqrt{n}} \Longrightarrow X_1^{(z)}.$$

Proof. Because the alphabets are finite, there exists $K < \infty$ such that every centered one-symbol surprisal increment has absolute value at most K . Hence

$$\sup_t |X_n(t) - X_n(t-)| \leq \frac{K}{\sqrt{n}} \rightarrow 0.$$

The processes X_n are martingales. Their predictable quadratic variations are

$$\langle X_n \rangle_t = \frac{1}{n} \sum_{k=1}^{\lfloor nt \rfloor} \sigma_z^2 \left(X_n \left(\frac{k-1}{n} \right) \right).$$

Consequently,

$$b^2 t + o(1) \leq \langle X_n \rangle_t \leq a^2 t + o(1).$$

The bounded-jump martingale tightness criterion therefore gives tightness of (X_n) . Every subsequential limit X is a continuous martingale.

It remains only to identify its quadratic variation. The coefficient σ_z^2 is bounded and has a single discontinuity, at z . Any subsequential limit has quadratic variation density bounded below by $b^2 > 0$. By the occupation-density formula,

$$\int_0^1 \mathbf{1}_{\{X_t=z\}} d\langle X \rangle_t = 0 \quad \text{a.s.}$$

Since

$$d\langle X \rangle_t \geq b^2 dt,$$

it follows that

$$\text{Leb}\{t : X_t = z\} = 0 \quad \text{a.s.}$$

The discontinuity of σ_z^2 is therefore encountered on a Lebesgue-null time set. The Riemann sums defining $\langle X_n \rangle$ pass to the limit and give

$$\langle X \rangle_t = \int_0^t \sigma_z^2(X_s) ds.$$

Hence every subsequential limit solves the martingale problem

$$X_t^2 - \int_0^t \sigma_z^2(X_s) ds \quad \text{is a martingale.}$$

Equivalently, it is a weak solution of

$$dX_t = \sigma_z(X_t) dB_t.$$

Because $0 < b \leq \sigma_z \leq a < \infty$ and the coefficient is piecewise constant with one interface, this one-dimensional diffusion has a unique weak law. Therefore all subsequences have the same limit, proving the claimed convergence. $\square$

1.3 The threshold envelope

Let

$$F_z(x) := \Pr(X_1^{(z)} \leq x).$$

After translating the interface from z to 0, the limiting process is the oscillating Brownian motion with diffusion coefficient a below the interface and b above it.

Set

$$\beta := \frac{a-b}{a+b}.$$

Lemma 1.4 (Pointwise threshold envelope). *For every $x \in \mathbb{R}$,*

$$\boxed{\inf_{z \in \mathbb{R}} F_z(x) = F_\star(x),}$$

where

$$F_\star(x) = \begin{cases} \frac{2b}{a+b} \Phi(x/b), & x \leq 0, \\ 1 - \frac{2a}{a+b} \Phi(-x/a), & x \geq 0. \end{cases}$$

Moreover the infimum is attained at

$$z = x.$$

The distribution $F_\star$ has mean

$$\boxed{\int_{\mathbb{R}} x dF_\star(x) = \sqrt{\frac{2}{\pi}}(a-b).}$$

Proof. The explicit oscillating-Brownian transition density gives the following elementary formulas.

For $x \geq 0$,

$$F_z(x) = \begin{cases} \Phi(x/b) - \beta \Phi((2z-x)/b), & z \leq 0, \\ -\beta + (1+\beta) \Phi\left(\frac{x}{b} + z\left(\frac{1}{a} - \frac{1}{b}\right)\right), & 0 \leq z \leq x, \\ \Phi(x/a) - \beta \Phi((x-2z)/a), & z \geq x. \end{cases}$$

Since $a > b$, the first two branches are decreasing in z , whereas the third is increasing. Thus the minimum occurs at $z = x$, giving

$$F_\star(x) = 1 - \frac{2a}{a+b} \Phi(-x/a), \quad x \geq 0.$$

For $x \leq 0$,

$$F_z(x) = \begin{cases} \Phi(x/b) - \beta \Phi((2z-x)/b), & z \leq x, \\ (1-\beta) \Phi\left(\frac{x}{a} + z\left(\frac{1}{b} - \frac{1}{a}\right)\right), & x \leq z \leq 0, \\ \Phi(x/a) - \beta \Phi((x-2z)/a), & z \geq 0. \end{cases}$$

The first branch decreases up to $z = x$, and the remaining branches increase from $z = x$. Hence

$$F_\star(x) = \frac{2b}{a+b} \Phi(x/b), \quad x \leq 0.$$

Differentiating,

$$dF_\star(x) = \begin{cases} \frac{2}{a+b} \phi(x/b) dx, & x < 0, \\ \frac{2}{a+b} \phi(x/a) dx, & x > 0. \end{cases}$$

Therefore

$$\begin{aligned} \mathbb{E}_{F_\star} X &= \frac{2}{a+b} \left[\int_{-\infty}^0 x \phi(x/b) dx + \int_0^\infty x \phi(x/a) dx \right] \\ &= \frac{2}{a+b} \frac{a^2 - b^2}{\sqrt{2\pi}} \\ &= \sqrt{\frac{2}{\pi}}(a-b). \end{aligned}$$

□

Theorem 1.5 (Adaptive causal converse). *For finite equal-entropy actions P, Q ,*

$$\liminf_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} \geq \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q|.$$

Proof. Assume $a = \sigma_P \geq b = \sigma_Q$. Fix a finite threshold set

$$\mathcal{Z} = \{z_1, \dots, z_r\}.$$

Apply Lemma 1.2 to the corresponding threshold policies:

$$H(S) - nh \geq \mathbb{E} \max_{z \in \mathcal{Z}} M_{z,n}.$$

Put

$$R_{z,n} := \frac{M_{z,n}}{\sqrt{n}}.$$

For the maximum

$$R_{\mathcal{Z},n} = \max_{z \in \mathcal{Z}} R_{z,n},$$

one always has

$$\Pr(R_{\mathcal{Z},n} \leq x) \leq \min_{z \in \mathcal{Z}} \Pr(R_{z,n} \leq x).$$

By Lemma 1.3, each marginal converges to F_z . The increments are uniformly bounded and all endpoint second moments are uniformly bounded, so the finite-family maxima are uniformly integrable. Hence

$$\liminf_{n \rightarrow \infty} \mathbb{E} R_{\mathcal{Z},n} \geq \mathbb{E} R_{\mathcal{Z}},$$

where $R_{\mathcal{Z}}$ has distribution function

$$x \mapsto \min_{z \in \mathcal{Z}} F_z(x).$$

Now take an increasing sequence of finite threshold grids whose union is dense in $\mathbb{R}$. By continuity of the explicit transition laws,

$$\min_{z \in \mathcal{Z}} F_z(x) \downarrow \inf_{z \in \mathbb{R}} F_z(x) = F_{\star}(x).$$

The limiting first moment is therefore

$$\sqrt{\frac{2}{\pi}}(a - b)$$

by Lemma 1.4.

Since the argument applies to every exact seed,

$$\liminf_{n \rightarrow \infty} \frac{C_n^{\text{causal}} - nh}{\sqrt{n}} \geq \sqrt{\frac{2}{\pi}}(a - b).$$

Interchanging P, Q if necessary gives the absolute value. □

1.4 Sorted arithmetic recycling

We now construct the matching upper bound.

Fix a block length L .

For every word $x \in \text{supp}(P)^L$, form an interval

$$J_P(x) \subset [0, 1)$$

of length

$$|J_P(x)| = P^L(x).$$

Arrange these intervals in nondecreasing order of

$$\iota_{P^L}(x) = -\log_2 P^L(x).$$

Ties may be resolved arbitrarily.

Thus all words having the same surprisal occupy one contiguous interval.

Construct analogously the ordered Q^L -partition

$$\{J_Q(w) : w \in \text{supp}(Q)^L\}.$$

Let

$$\alpha_P(x), \quad \alpha_Q(w)$$

denote the corresponding left endpoints.

Take

$$U_0 \sim \text{Unif}[0, 1)$$

and independently

$$W_1, W_2, \dots \stackrel{\text{iid}}{\sim} Q^L.$$

Recursively define X_j, V_j, U_j by

$$\begin{aligned} U_{j-1} &\in J_P(X_j), \\ V_j &:= \frac{U_{j-1} - \alpha_P(X_j)}{P^L(X_j)}, \end{aligned}$$

and

$$\boxed{U_j = \alpha_Q(W_j) + Q^L(W_j)V_j.}$$

We call this the *sorted arithmetic swap*.

Lemma 1.6 (Exact recycling invariant). *For every $j \geq 1$,*

$$U_j \sim \text{Unif}[0, 1),$$

and

$$X_1, X_2, \dots \stackrel{\text{iid}}{\sim} P^L.$$

Moreover

$$X_j \in \sigma(U_0, W_1, \dots, W_{j-1}),$$

so the block coupling is strictly one-way causal.

Proof. Suppose U_{j-1} is uniform and independent of $X_1, \dots, X_{j-1}$.

Arithmetic decomposition of the P^L -partition gives

$$X_j \sim P^L, \quad V_j \sim \text{Unif}[0, 1),$$

with V_j independent of X_j and of the preceding X -blocks.

The block W_j is fresh and independent with law Q^L . Conditional on $W_j = w$,

$$U_j = \alpha_Q(w) + Q^L(w)V_j$$

is uniform on $J_Q(w)$. Mixing with weights $Q^L(w)$ makes U_j uniform on all of $[0, 1)$.

The pair (W_j, V_j) is independent of $X_1, \dots, X_j$, so U_j is also independent of those blocks. Induction proves the product law.

Finally X_j is determined before W_j is used, and therefore depends only on $U_0, W_1, \dots, W_{j-1}$. $\square$

1.5 Exact monotone coupling of block surprisal

Define

$$A_j := \iota_{P^L}(X_j) - Lh, \quad B_j := \iota_{Q^L}(W_j) - Lh,$$

and

$$Z_j := A_{j+1} - B_j.$$

Lemma 1.7 (Monotone block-spectrum coupling). *For every j , the pair*

$$(B_j, A_{j+1})$$

is exactly the monotone quantile coupling of the centered block-surprisal laws.

Consequently,

$$\mathbb{E}Z_j = 0$$

and

$$\frac{\mathbb{E}Z_j^2}{L} = W_2^2 \left(\text{Law} \frac{A_L}{\sqrt{L}}, \text{Law} \frac{B_L}{\sqrt{L}} \right).$$

Moreover,

$$\boxed{\frac{\mathbb{E}Z_j^2}{L} \longrightarrow (\sigma_P - \sigma_Q)^2.}$$

Proof. Because the Q^L -intervals are arranged by surprisal,

$$B_j = F_{Q,L}^{-1}(U_j)$$

almost surely, where $F_{Q,L}$ is the distribution function of centered Q^L -surprisal.

Likewise,

$$A_{j+1} = F_{P,L}^{-1}(U_j).$$

Thus both variables are increasing functions of the same uniform coordinate U_j , which is precisely the monotone coupling.

The one-dimensional monotone coupling is optimal for quadratic transport, hence

$$\mathbb{E}Z_j^2 = W_2^2(\text{Law}(A_L), \text{Law}(B_L)).$$

By the central limit theorem,

$$\frac{A_L}{\sqrt{L}} \Longrightarrow N(0, V_P), \quad \frac{B_L}{\sqrt{L}} \Longrightarrow N(0, V_Q).$$

Their second moments are exactly V_P, V_Q , so the convergence is also in W_2 . Therefore

$$W_2 \left(\text{Law} \frac{A_L}{\sqrt{L}}, \text{Law} \frac{B_L}{\sqrt{L}} \right) \rightarrow W_2(N(0, V_P), N(0, V_Q)).$$

For centered one-dimensional Gaussians,

$$W_2(N(0, V_P), N(0, V_Q)) = |\sigma_P - \sigma_Q|.$$

□

1.6 The dependence defect is exponentially rare

The variables Z_j have the correct one-block law but are not automatically independent because the interval state U_j is recycled.

The next lemma shows that this dependence is invisible at second order.

Let

$$p_\star := \max_x P(x) < 1.$$

Let $N_Q(L)$ be the number of distinct surprisal values of Q^L . If the support of Q has d_Q points, then

$$N_Q(L) \leq (L + 1)^{d_Q - 1},$$

because block surprisal is determined by the empirical type.

Call a P^L -atom $J_P(x)$ *bad* if it contains a boundary between two consecutive Q^L -surprisal levels.

Lemma 1.8 (Exponential boundary localization). *The bad set satisfies*

$$P^L\{x : J_P(x) \text{ is bad}\} \leq N_Q(L)p_\star^L.$$

On the complement of the bad set, B_j is a deterministic function of X_{j+1} .

Proof. There are at most $N_Q(L) - 1$ internal boundaries between consecutive Q^L -surprisal levels.

Each boundary lies in at most one P^L -atom. Every P^L -atom has length at most

$$p_\star^L.$$

The total Lebesgue measure of the union of all bad atoms is therefore at most

$$N_Q(L)p_\star^L.$$

Since $X_{j+1} \sim P^L$, this Lebesgue measure is exactly the probability of the bad event.

If $J_P(X_{j+1})$ is good, it lies entirely inside a single Q^L -surprisal level. Hence the value of B_j , which is constant on that level, is determined by X_{j+1} alone. □

Choose a function G_L on P^L -words so that on good atoms

$$G_L(X_{j+1}) = Z_j.$$

Define it arbitrarily on bad atoms and write

$$Y_{L,j} := G_L(X_{j+1}).$$

By Lemma 1.6, the variables

$$Y_{L,1}, Y_{L,2}, \dots$$

are iid for fixed L .

Since the alphabets are finite, there exists $C < \infty$ such that

$$|A_j| + |B_j| + |Z_j| + |Y_{L,j}| \leq CL.$$

Therefore

$$\mathbb{E}|Z_j - Y_{L,j}| \leq CL N_Q(L) p_\star^L,$$

and

$$\mathbb{E}|Z_j - Y_{L,j}|^2 \leq C^2 L^2 N_Q(L) p_\star^L.$$

In particular,

$$\frac{\text{Var}(Y_{L,j})}{L} \longrightarrow (\sigma_P - \sigma_Q)^2.$$

1.7 The running information deficit

For horizon n , choose

$$L = L_n := \lceil (\log_2 n)^2 \rceil,$$

and

$$m := \left\lfloor \frac{n}{L} \right\rfloor.$$

Then

$$L \rightarrow \infty, \quad L = o(\sqrt{n}),$$

and, for every fixed nondegenerate P ,

$$n \text{ poly}(L) p_\star^L = o(1).$$

Let

$$\tilde{Y}_{L,j} := Y_{L,j} - \mathbb{E}Y_{L,j}.$$

The triangular array

$$\{\tilde{Y}_{L_n,j} : 1 \leq j < m\}$$

consists of row-wise iid centered variables satisfying

$$|\tilde{Y}_{L,j}| \leq CL = o(\sqrt{n}),$$

and

$$\frac{m \text{Var}(\tilde{Y}_{L,1})}{n} \longrightarrow (\sigma_P - \sigma_Q)^2.$$

Thus the Lindeberg condition holds automatically and the functional central limit theorem gives

$$\left(\frac{1}{\sqrt{n}} \sum_{j=1}^{\lfloor mt \rfloor} \tilde{Y}_{L,j} \right)_{0 \leq t \leq 1} \Longrightarrow (|\sigma_P - \sigma_Q| B_t)_{0 \leq t \leq 1}.$$

The maximal L^2 inequality gives uniform integrability of the maxima. Hence

$$\frac{1}{\sqrt{n}} \mathbb{E} \max_{0 \leq k < m} \sum_{j=1}^k \tilde{Y}_{L,j} \longrightarrow |\sigma_P - \sigma_Q| \mathbb{E} \sup_{0 \leq t \leq 1} B_t.$$

By the reflection principle,

$$\mathbb{E} \sup_{0 \leq t \leq 1} B_t = \sqrt{\frac{2}{\pi}}.$$

The boundary estimates above imply

$$\mathbb{E} \max_{0 \leq k < m} \left| \sum_{j=1}^k (Z_j - Y_{L,j}) \right| = o(\sqrt{n}),$$

and

$$m|\mathbb{E}Y_{L,1}| = o(\sqrt{n}).$$

Therefore

$$\boxed{\mathbb{E} \max_{0 \leq k < m} \sum_{j=1}^k Z_j = \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| \sqrt{n} + o(\sqrt{n}).} \quad (*)$$

1.8 Finite-bit exactization of the interval state

The uniform coordinate U_0 is only a convenient representation. The causal source cost is Shannon entropy, so the continuous coordinate must not be counted as a free infinite random tape.

We now replace it by a finite prefix-free fair-bit transcript.

For each block j , define

$$d_j := \min\{V_j, 1 - V_j\}.$$

The distance from U_{j-1} to the nearer boundary of its current P^L -atom is

$$\delta_j = P^L(X_j)d_j.$$

For a fixed realization $W_1, \dots, W_m$, reveal the binary expansion of U_0 until the current dyadic cylinder determines $X_1, \dots, X_m$ uniquely.

Let B_m be the number of revealed fair bits.

Lemma 1.9 (Prefix exactization bound). *Almost surely,*

$$B_m \leq Lh + A_1 + \max_{0 \leq k < m} \sum_{j=1}^k Z_j + G_m + 2,$$

where

$$G_m := \max_{1 \leq j \leq m} \log_2 \frac{1}{d_j}.$$

Furthermore,

$$\mathbb{E}G_m \leq \log_2(2m) + \frac{1}{\ln 2} + 1.$$

Consequently,

$$\boxed{H(X_1, \dots, X_m \mid W_1, \dots, W_m) \leq Lh + \mathbb{E} \max_{0 \leq k < m} \sum_{j=1}^k Z_j + O(\log m).}$$

Proof. Suppose b initial fair bits have been revealed. The corresponding dyadic interval has length 2^{-b} .

As long as the same output blocks are determined, the arithmetic swap is affine. At stage j , the image of this dyadic interval therefore has length

$$2^{-b} \prod_{i < j} \frac{Q^L(W_i)}{P^L(X_i)}.$$

A sufficient condition for this image to remain inside the realized P^L -atom is

$$2^{-b} \prod_{i < j} \frac{Q^L(W_i)}{P^L(X_i)} < \delta_j.$$

Taking $-\log_2$ gives

$$b > \sum_{i < j} [\imath_{P^L}(X_i) - \imath_{Q^L}(W_i)] + \imath_{P^L}(X_j) + \log_2 \frac{1}{d_j}.$$

Using

$$A_i = \imath_{P^L}(X_i) - Lh, \quad B_i = \imath_{Q^L}(W_i) - Lh,$$

the first two terms telescope exactly:

$$\sum_{i < j} (A_i - B_i) + Lh + A_j = Lh + A_1 + \sum_{i=1}^{j-1} (A_{i+1} - B_i).$$

Therefore

$$B_m \leq Lh + A_1 + \max_{0 \leq k < m} \sum_{i=1}^k Z_i + G_m + 2.$$

By Lemma 1.6, V_j is uniform on $[0, 1)$. Hence for $0 < \varepsilon \leq 1/2$,

$$\Pr(d_j \leq \varepsilon) = 2\varepsilon.$$

A union bound gives

$$\Pr(G_m > s) \leq \min\{1, 2m 2^{-s}\}.$$

Integration of this tail yields

$$\mathbb{E}G_m \leq \log_2(2m) + \frac{1}{\ln 2} + 1.$$

For fixed $W_1, \dots, W_m$, the terminal fair-bit strings form a prefix-free complete family outside a null set of interval boundaries. If R denotes the terminal bit string, then

$$\Pr(R = r \mid W_1, \dots, W_m) = 2^{-|r|}.$$

Thus

$$H(R \mid W_1, \dots, W_m) = \mathbb{E}[B_m \mid W_1, \dots, W_m].$$

The X -blocks are deterministic functions of R and the W -blocks, so

$$H(X_1, \dots, X_m \mid W_1, \dots, W_m) \leq H(R \mid W_1, \dots, W_m).$$

Average over W and use

$$\mathbb{E}A_1 = 0.$$

□

1.9 From a causal stream to a causal response tree

For completeness, we restate the stream-to-tree lift in the form used here.

Lemma 1.10 (Anti-diagonal stream-to-tree lift). *Let*

$$W^N \sim Q^N, \quad X^N \sim P^N,$$

and suppose

$$\text{Law}(X_{1:k} \mid W_{1:N}) = \text{Law}(X_{1:k} \mid W_{1:k})$$

for every $k \leq N$.

Then there exists an exact horizon- N C3-C response tree with source entropy at most

$$H(X^N, W^N).$$

In particular,

$$C_N^{\text{causal}}(P, Q) \leq H(X^N, W^N).$$

Proof. Reverse the W -stream:

$$\widehat{W}_s := W_{N-s+1}.$$

On the r -th occurrence of action P , output X_r . On the s -th occurrence of action Q , output $\widehat{W}_s$.

Consider a deterministic action word containing p occurrences of P and q occurrences of Q , with

$$p + q \leq N.$$

The path exposes

$$X_{1:p}$$

and

$$W_{N-q+1:N}.$$

Since

$$N - q + 1 \geq p + 1,$$

the exposed W -coordinates lie strictly after the W -prefix on which $X_{1:p}$ may depend.

By the one-way causal assumption and the product law of W ,

$$X_{1:p} \perp W_{N-q+1:N}.$$

Their laws are respectively

$$P^p \quad \text{and} \quad Q^q.$$

Hence every deterministic action word has exactly the required product output law.

The action-tree characterization then gives exactness for every adaptive nonanticipative policy. $\square$

1.10 The matching constructive upper bound

Theorem 1.11 (Sharp constructive causal upper bound). *For finite nondegenerate P, Q with*

$$H(P) = H(Q) = h,$$

$$\limsup_{n \rightarrow \infty} \frac{C_n^{\text{causal}}(P, Q) - nh}{\sqrt{n}} \leq \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q|.$$

Proof. Let

$$L = \lceil (\log_2 n)^2 \rceil, \quad m = \left\lfloor \frac{n}{L} \right\rfloor, \quad N = mL.$$

Apply the sorted arithmetic swap to m blocks. By Lemma 1.6, after flattening the blocks,

$$X^N \sim P^N, \quad W^N \sim Q^N,$$

and the coupling is one-way causal.

Since

$$H(W^N) = Nh,$$

Lemma 1.9 and (*) give

$$\begin{aligned} H(X^N, W^N) &= Nh + H(X^N | W^N) \\ &\leq Nh + \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| \sqrt{N} + o(\sqrt{N}). \end{aligned}$$

Lemma 1.10 therefore yields

$$C_N^{\text{causal}}(P, Q) \leq Nh + \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| \sqrt{N} + o(\sqrt{N}).$$

There remain fewer than L positions. Generate these with an independent exact finite causal realization. Its entropy cost is $O(L)$, and

$$L = O((\log n)^2) = o(\sqrt{n}).$$

Also

$$N = n - O(L), \quad \sqrt{N} = \sqrt{n} + o(\sqrt{n}).$$

Therefore

$$C_n^{\text{causal}}(P, Q) \leq nh + \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| \sqrt{n} + o(\sqrt{n}).$$

□

1.11 Proof of the sharp law

Proof of Theorem 1.1. The adaptive converse, Theorem 1.5, gives

$$\liminf_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} \geq \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q|.$$

The sorted-arithmetic construction, Theorem 1.11, gives

$$\limsup_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} \leq \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q|.$$

The two bounds coincide.

□

1.12 The crash-test pair

For the crash-test pair

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$$

and

$$Q = (q, q, 1 - 2q),$$

where q is chosen so that

$$H(Q) = H(P) = 1.5,$$

one has

$$q = 0.41003242818198557608 \dots$$

The varentropies are

$$V_P = \frac{1}{4},$$

and

$$V_Q = 0.208347310239073 \dots,$$

so

$$\sigma_P = 0.5, \quad \sigma_Q = 0.456450775263963 \dots$$

Hence

$$\kappa_{\text{crash}} := \sqrt{\frac{2}{\pi}}(\sigma_P - \sigma_Q) = 0.034747254051818 \dots$$

Corollary 1.12 (Exact crash-test second order). *For the crash-test pair,*

$$C_n^{\text{causal}} = 1.5n + 0.034747254051818 \dots \sqrt{n} + o(\sqrt{n}).$$

Equivalently,

$$\lim_{n \rightarrow \infty} \frac{C_n^{\text{causal}} - 1.5n}{\sqrt{n}} = 0.034747254051818 \dots$$

Thus the finite-horizon causal residual is neither bounded nor logarithmic:

$$E_n = \Theta(\sqrt{n}).$$

More precisely,

$$E_n = 0.034747254051818 \dots \sqrt{n} + o(\sqrt{n}).$$

At the same time,

$$\frac{E_n}{n} \rightarrow 0,$$

so the first-order causal Shannon floor remains exactly

$$C_\infty^{\text{causal}} = 1.5.$$

1.13 What the coefficient means

The coefficient has a direct information-accounting interpretation.

Both actions produce the same mean information,

$$h.$$

The linear mismatch therefore cancels.

The sorted arithmetic swap couples the two block-surprisal processes by the same asymptotic Gaussian coordinate:

$$A_L \simeq \sigma_P \sqrt{L} Z, \quad B_L \simeq \sigma_Q \sqrt{L} Z.$$

Their residual mismatch is therefore

$$A_L - B_L \simeq (\sigma_P - \sigma_Q) \sqrt{L} Z.$$

Causality cannot borrow information from the future. The source must cover the largest prefix deficit, which becomes in the diffusion limit

$$|\sigma_P - \sigma_Q| \sup_{0 \leq t \leq 1} B_t.$$

Since

$$\mathbb{E} \sup_{0 \leq t \leq 1} B_t = \sqrt{\frac{2}{\pi}},$$

the exact second-order bill is

$$\boxed{\sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q|}.$$

The $\sqrt{n}$ term is therefore not a discretization artifact. It is the causal price of unequal information fluctuations after the common mean Shannon rate has been removed.

1.14 Status of the previous open problems

The finite causal residual problem is now resolved at second order.

For equal-entropy finite two-action channels,

$$\boxed{E_n = \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| \sqrt{n} + o(\sqrt{n})}.$$

In particular, when

$$V_P \neq V_Q,$$

the residual has the exact order

$$\Theta(\sqrt{n}).$$

When

$$V_P = V_Q,$$

the theorem gives

$$E_n = o(\sqrt{n}),$$

and the next nonvanishing asymptotic scale remains a separate higher-order problem.

For the crash-test pair the second-order problem is completely closed. The only remaining finite-horizon asymptotic question is the *third-order* structure of

$$E_n - 0.034747254051818 \dots \sqrt{n}.$$

1.15 Relation to neighboring coding results

The construction should not be confused with ordinary independent block coupling.

Independent block couplings pay their spectrum mismatch separately in each block and therefore do not preserve the sharp running-deficit constant.

The sorted arithmetic swap instead recycles the same relative interval coordinate from one block to the next. This simultaneously gives

exact target marginals,

one-way causality,

monotone coupling of block information spectra,

and

only $o(\sqrt{n})$ consistency overhead.

The mechanism is closely related in spirit to arithmetic random-number conversion and randomness recycling, but the statement proved here is a source-to-source causal coupling theorem tailored to the C3-C response-tree cost.

The informational-variance obstruction is consistent with the finitary coding phenomenon that equal entropy alone does not remove the critical effect of unequal information variances.

J. Keilson and J. A. Wellner, *Oscillating Brownian Motion*, Journal of Applied Probability 15(2) (1978), 300–310.

A. Lejay and P. Pigato, *Statistical Estimation of the Oscillating Brownian Motion*, Bernoulli 24(4B) (2018), 3568–3602. <https://arxiv.org/abs/1701.02129>